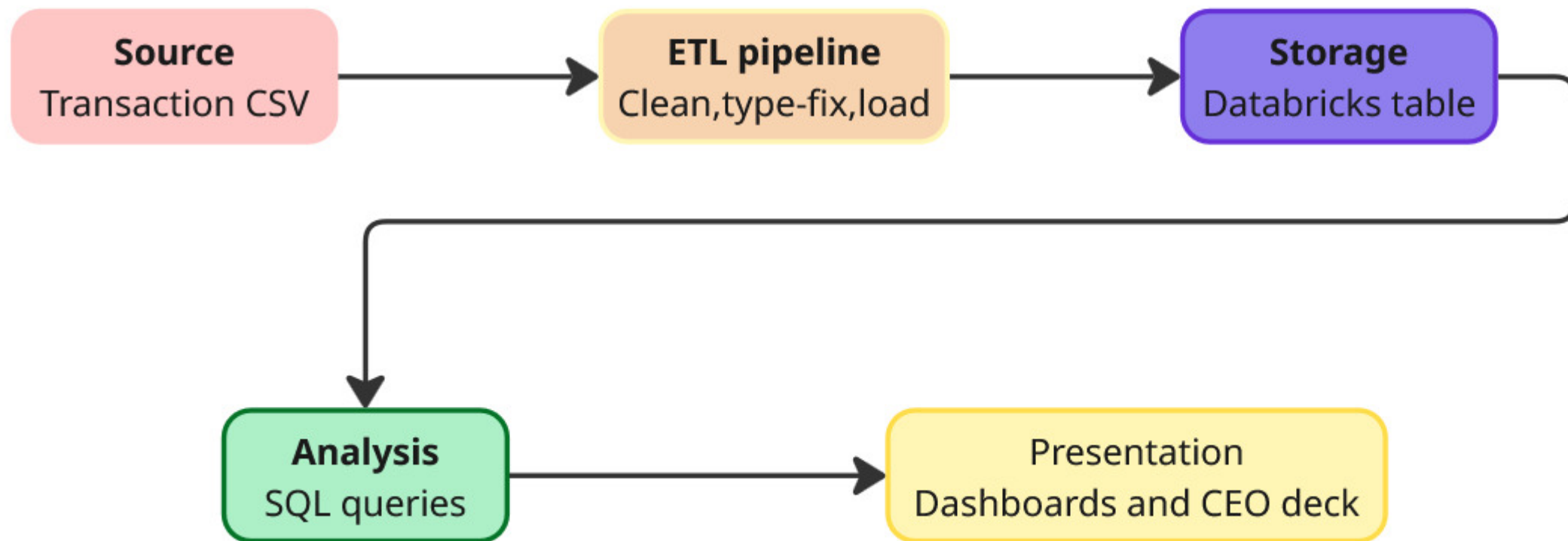


MATENCHI KAGISO

Bright Coffee Shop Sales Analysis — Miro Project Plan



Step 0: Read & understand the problem statement

Bright Coffee Shop's new CEO wants to grow revenue and improve product performance. As Junior Data Analyst, prepare a presentation covering:

- Which products generate the most revenue
- What time of day the store performs best
- Sales trends across products and time intervals
- Recommendations for improving sales performance

Notes:

- Dataset: transaction_id, transaction_date, transaction_time, transaction_qty, store_id, store_location, product_id, unit_price, product_category, product_type, product_detail
- ~149,000 transaction rows
- unit_price uses comma decimals (e.g. "3,1") — flag for cleaning
- No revenue/total_amount column exists yet — must be calculated

Step 1: Define your analytical framework

i) Understand the business requirement: grow revenue, improve product performance

ii) Identify what metrics matter: revenue, units sold, revenue per time slot, product mix

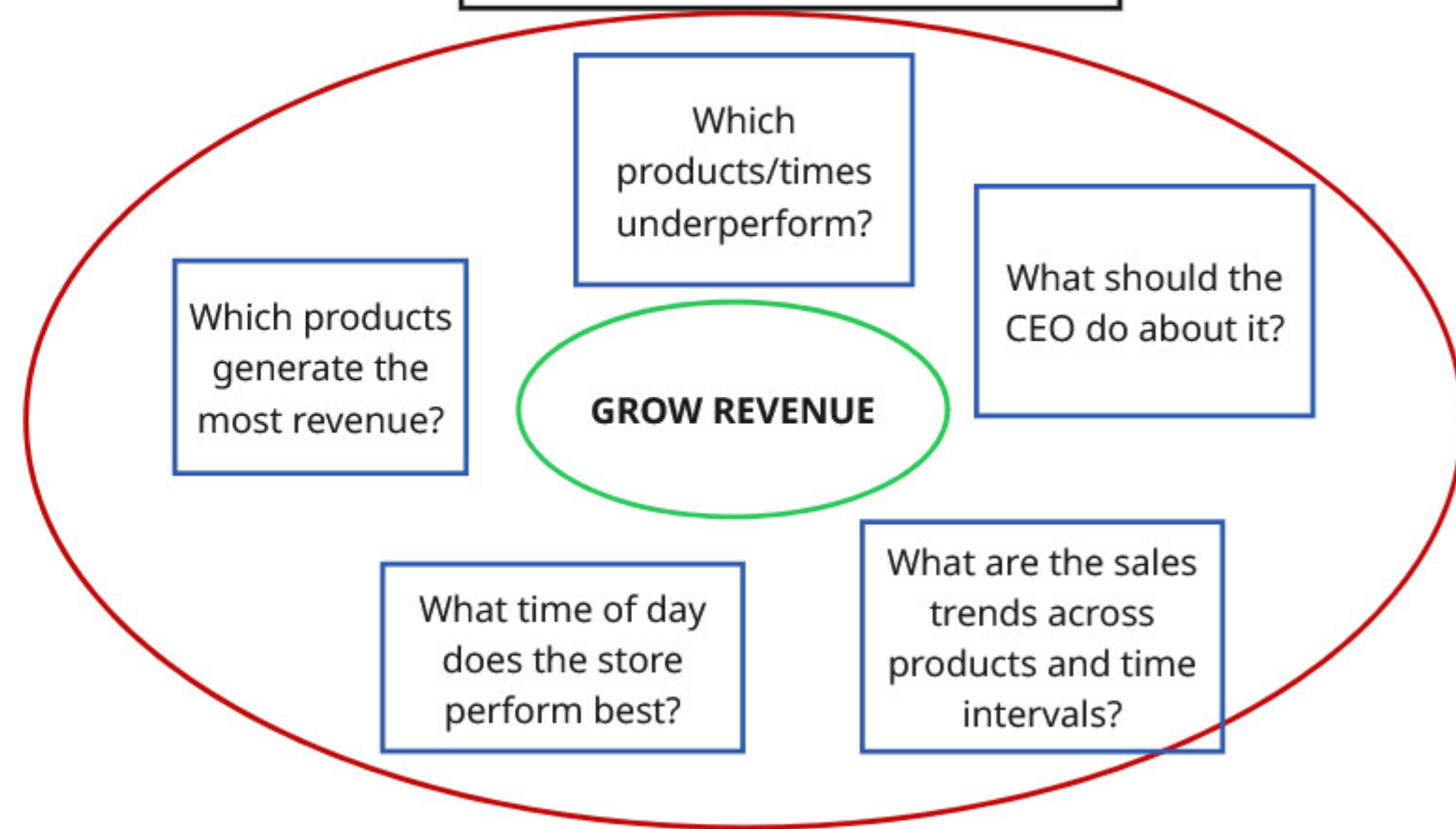
iii) Map which data columns answer which questions

Revenue by product → unit_price, transaction_qty, product_category/type/detail
Time-of-day performance → transaction_time
Trends over time → transaction_date
Store comparisons (bonus) → store_id, store_location

Step 3: Data architecture & pipeline

Source → Raw transactions CSV **ETL pipeline**
→ Clean, fix decimal formatting, cast types, remove duplicates, load **Storage** → Databricks table (single source of truth for analysis)
Analysis → SQL queries against the Databricks table **Presentation** → Charts, insights, and CEO-facing deck

Step 2: Answer data-driven questions



Step 4: Data exploration

I) Check data shape, columns, and data types

II) Identify missing or null values

III) Check date ranges (confirm full period covered, look

IV) Confirm unit_price is stored as text/comma-decimal or if it needs

V) Create derived columns:
Revenue per transaction (see Step 6 calculations)
Hour of day / 30-minute time bucket

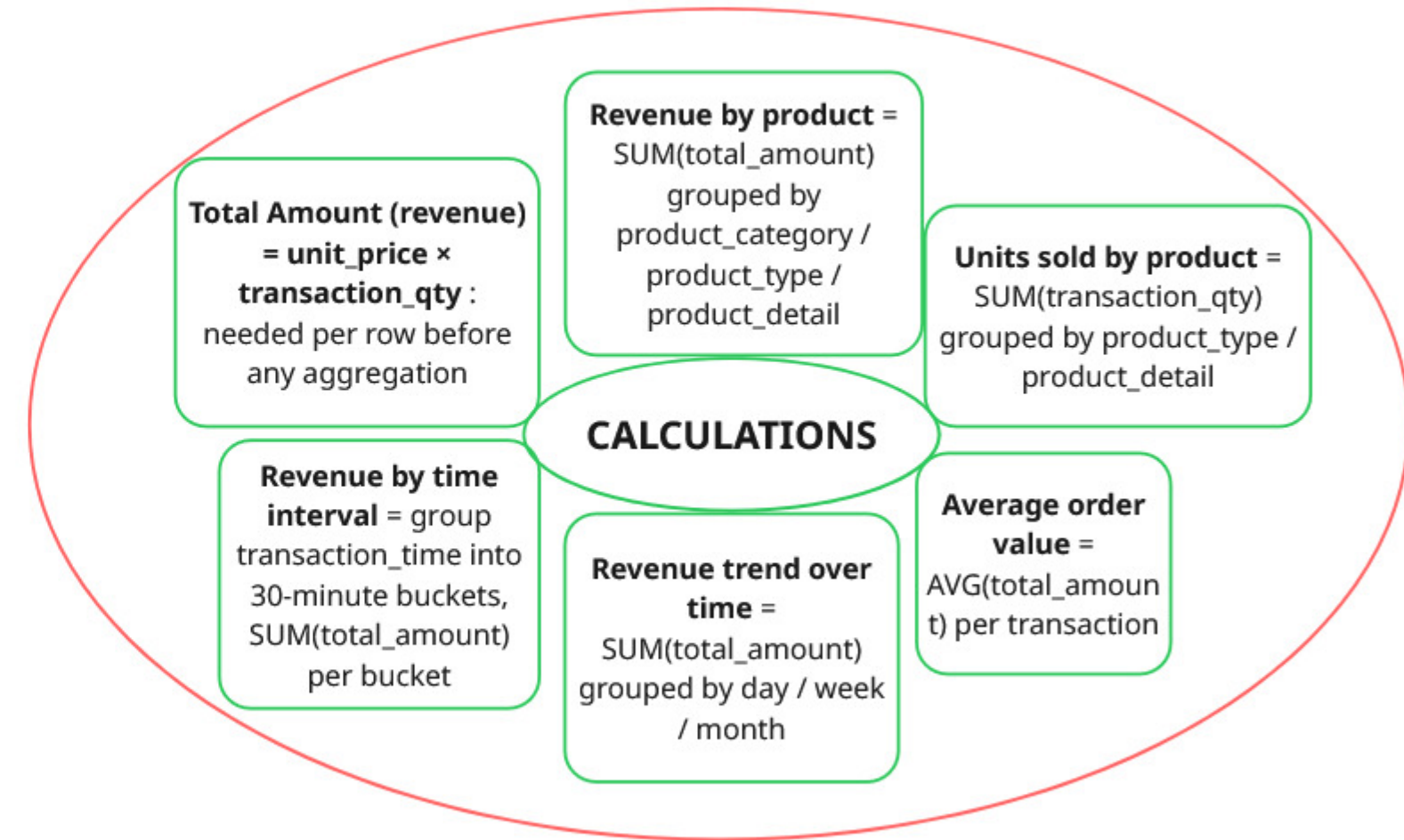
Step 5: Data cleaning

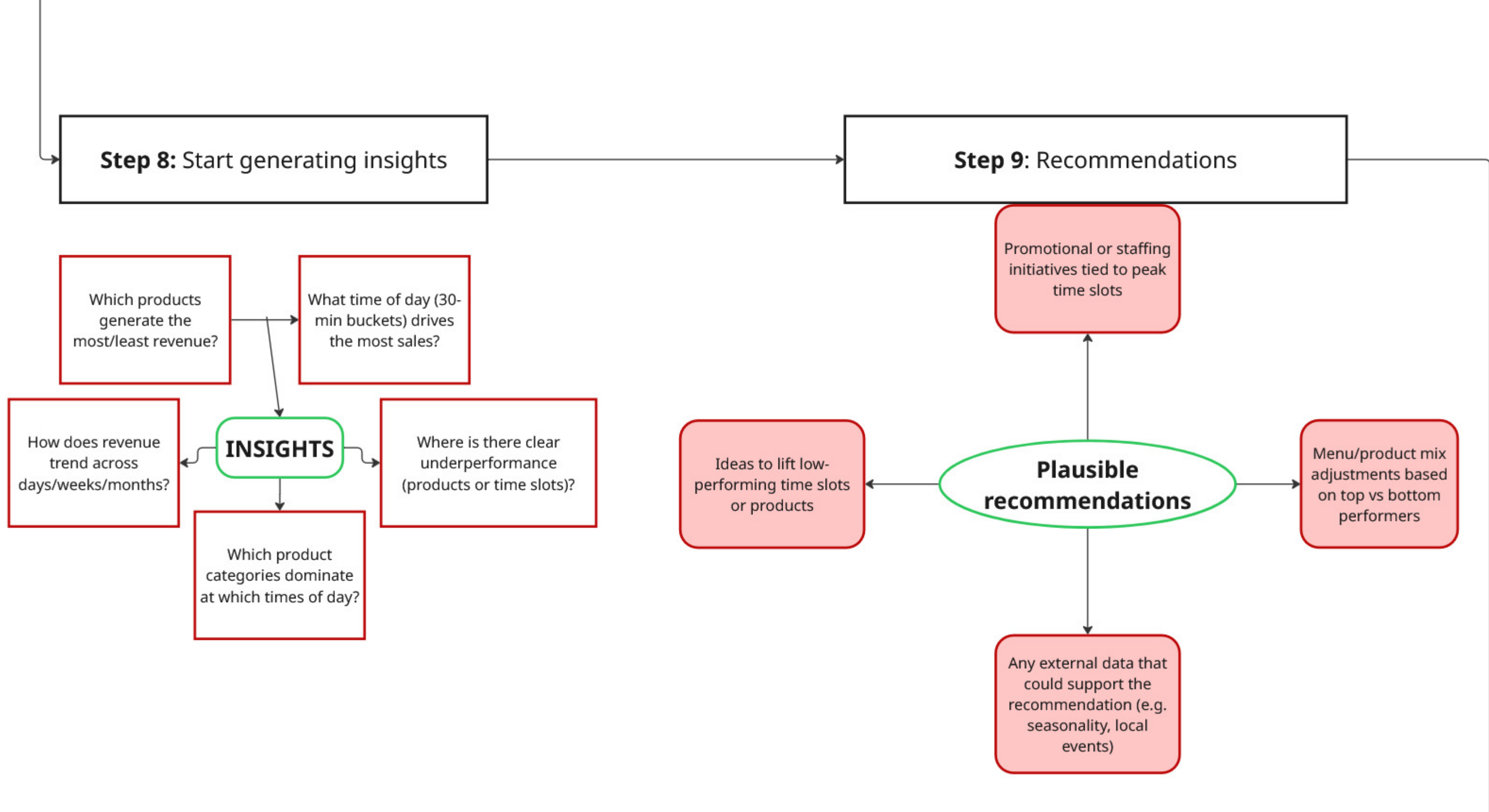
- i) Convert unit_price from comma-decimal string to proper numeric (e.g. "3,1" → 3.1)
- ii) Check for duplicate transaction_ids
- iii) Standardize text fields (product_category, product_type, product_detail, store_location) — watch for inconsistent casing or near-duplicate names
- iv) Handle any missing/null values in qty, price, or product fields
- v) Validate product_id is consistent with product_category/type/detail (no mismatches)

Step 7: Data analysis

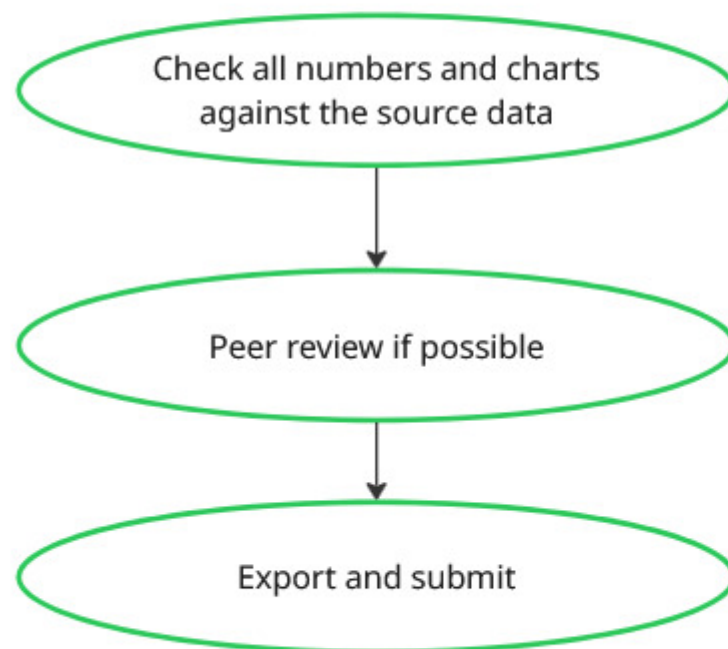
- i) Revenue by product category/type/detail (top and bottom performers)
- ii) Revenue and transaction volume by 30-minute time interval — identify peak windows
- iii) Sales trends over time (daily/weekly/monthly)
- iv) Product mix by time of day (does morning skew coffee, afternoon skew tea/pastries?)
- v) (Bonus) Store location comparison

Step 6: Outline the calculations to be performed





Step 11: Review & finalize



Step 10: Build the presentation

- *Build charts/visuals for each insight
- *Gather talking points per insight
- *Anticipate likely CEO questions (e.g. "why is this product underperforming?", "what would it take to shift revenue into slow hours?")